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# Present at the Muster, Missing from the Roster

*A Term-Long Audit of a Fire-Drill Assembly Headcount Against a Building's Own Swipe-Card Entry Log, and the Floor-Warden Roster It Still Sizes*

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**Abstract.** A nine-storey administration building's fire-safety framework recalculates each floor's warden establishment once a term from the trailing mean of that floor's fire-drill assembly-point headcount, on the reasoning that the figure marshals hand in within the hour is the only occupancy reading Facilities and Safety has in time to act on. We test whether that headcount agrees with the number the building's own swipe-card system would report for the same alarm, using entry and exit records in the ten minutes before each drill's alarm across seven drills and nine floors over one teaching term ($n = 63$ floor-drill pairs). The two readings correlate only moderately ($r = 0.62$), and the disagreement is not noise: the building's lowest three floors run high (a mean overcount of 16.0%, consistent with an adjacent uninstrumented annex's occupants joining the nearest assembly zone) and its highest three run low (a mean undercount of 14.0%, consistent with marshals finalising the tally before the longest stairwell descents complete), while the middle band sits closest to parity on average but carries the widest spread of the three. A marshal refresher briefing introduced after the fourth drill leaves the disagreement statistically unchanged (Mann-Whitney $p = 0.26$). Recalculating each floor's warden establishment from the swipe-derived mean instead of the assembly count changes the assigned establishment on three of the building's nine floors. We discuss what these results may suggest for occupancy proxies drawn from compliance procedures more broadly.

## Introduction

Fire-drill assembly points exist to answer one question — has everyone left the building — and the headcount marshals take there has, at the administration building we study, acquired a second job. Facilities and Safety's fire-safety framework treats the assembly count as a proxy for how many people a floor typically holds, and recalculates that floor's warden establishment each term from the trailing average of its own drills. The appeal is practical: the tally is complete and legible within the hour a drill finishes, whereas the building's swipe-card system — which logs every entry and exit against a floor and could, in principle, answer the same occupancy question — takes a standing data request and several working days to return a comparable figure. An indicator adopted for its availability rather than its accuracy is a familiar shape at this institution [1], but it has not, to our knowledge, been tested directly against the ground truth it stands in for at the point where it drives a staffing decision.

We ask whether the assembly-point headcount agrees with the occupancy the same floor's swipe-card log would report for the same alarm, and whether any disagreement is patterned enough to matter for the establishment the headcount sets.

We make three contributions, each hedged to the evidence available. First, we assemble a term-long, floor-matched record pairing each drill's assembly headcount against an independently derived swipe-card occupancy figure for the same ten-minute pre-alarm window, across seven drills and nine floors. Second, we show the two readings correlate only moderately overall, and that the residual disagreement follows a consistent floor-banded bias rather than reading as noise. Third, we test whether a marshal training intervention introduced partway through the term narrows that bias, and find it does not, and we discuss what a validated but retained proxy of this kind may suggest for occupancy-linked staffing decisions drawn from safety-compliance procedures more broadly.

## Related work

Occupancy estimation from passive infrastructure data is well established: badge and access-control logs, Wi-Fi sensing, and audio-or sensor-fusion approaches have each been used to infer how many people occupy a space without deploying a dedicated headcount [2], [3], [4], [5]. This literature treats accurate occupancy as the target signal and manual counting as a fallback or validation step, the inverse of the present setting, in which a manual count is the operational figure and a passive log sits unused as the closer approximation to ground truth.

Evacuation research has separately documented that occupant behaviour during a drill is uneven across a building: response delays, imbalanced exit use, and drill-specific complacency effects are all reported in recent studies of workplace and campus evacuations [6], [7], [8], any of which would bias a manual assembly count in ways a passive access log would not share.

The gap between a measure and the outcome it is meant to track has its own literature outside safety engineering, from Goodhart's law's original formulation and its recent formal variants [9], [10], [11], [12], [13] to empirical accounts of targets gamed or drifted from their intended referent in public-sector and university performance management [14], [15]. Prior work at this institution has found a related pattern locally: an indicator built to flag stale registry entries for retirement rarely produced retirements once its own uptime metric entered the Living Dashboard [1], and a capacity-setting occupancy sensor elsewhere on campus correlated only moderately with an independent manual headcount of the object it purported to count [16]. We extend this line of inquiry to a headcount that is itself the manual reading, tested here against the passive log it has stood in front of for years.

## Method

The study site is a nine-storey administration building whose nine occupied levels (Levels 1-9) each hold one to three open-plan work areas and share two stair cores and one lift bank. The building's swipe-card system logs a badge event (entry or exit) at each floor's card reader with a timestamp accurate to the second; its assembly procedure sends Levels 1-3 to Zone A (the north lawn), Levels 4-6 to Zone B (the forecourt), and Levels 7-9 to Zone C (the south car park), each zone worked by two rostered marshals who tally heads against a printed floor list and radio the total to Facilities and Safety within, per policy, fifteen minutes of the all-clear.

We recorded the marshal-reported headcount for every floor at seven fire drills (four scheduled, three unannounced) spread across one thirteen-week teaching term. For each drill, we separately queried the swipe-card system for every entry and exit event on each floor in the ten minutes preceding the alarm, and computed a swipe-derived occupancy figure as the running balance of entries minus exits since the floor's first badge event that day, frozen at the alarm timestamp. This yields $n = 63$ floor-drill pairs (seven drills across nine floors).

Facilities and Safety's fire-safety framework recalculates each floor's warden establishment once a term from the trailing three-drill mean of that floor's assembly headcount,

$$w_f = \left\lceil \frac{\overline{H}_f}{12} \right\rceil,$$

assigning one rostered warden per twelve people in the mean $\overline{H}_f$ and rounding up to the next whole warden, with no adjustment for a floor whose headcount and swipe-derived occupancy disagree by any margin. We compute, for comparison only, what establishment the same formula would assign if $\overline{H}_f$ were replaced by the trailing three-drill mean of the swipe-derived occupancy figure instead.

We report the Pearson correlation between the two occupancy readings across all 63 pairs, and decompose the mean signed disagreement (assembly headcount minus swipe figure, as a percentage of the swipe figure) by floor band (Levels 1-3, 4-6, 7-9). Beginning at the fifth drill, marshals received a fifteen-minute refresher briefing on tally procedure before the drill; we compare the floor-banded absolute bias before and after the briefing (drills 1-4 against drills 5-7, a Mann-Whitney U test given the ticket-level distributions are not assumed normal) as an ablation on whether training, rather than the zone-and-stairwell structure itself, drives the disagreement.

## Results

Across the 63 floor-drill pairs, marshal-reported headcounts correlate only moderately with the swipe-derived occupancy figure for the same alarm ($r = 0.62$, $n = 63$; Figure 1). Zone A (Levels 1-3) returns the highest headcounts relative to its swipe-derived figure, with a mean signed overcount of 16.0%; Zone C (Levels 7-9) shows the opposite pattern, a mean undercount of 14.0%; Zone B (Levels 4-6) sits closest to parity on average, a mean bias of 3.2%, though its spread is the widest of the three (Figure 2). Zone A's overcount is consistent with informal reports that occupants of an adjacent, uninstrumented annex use the north lawn as their own assembly point during a drill and are folded into the nearest marshal's tally; Zone C's undercount is consistent with the south car park marshals finalising and radioing a total before occupants descending from Level 9, the building's longest stair run, have fully arrived.

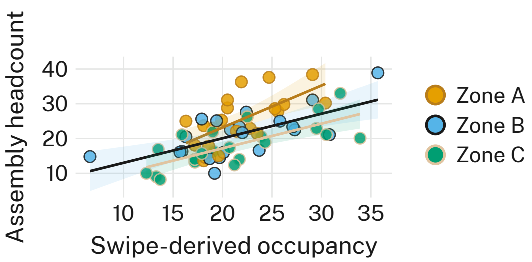


Figure 1: Swipe-derived occupancy against the same alarm's assembly headcount, per floor per drill ($n = 63$): the two readings correlate only moderately ($r = 0.62$), with Zone A running above the fitted line and Zone C below it.

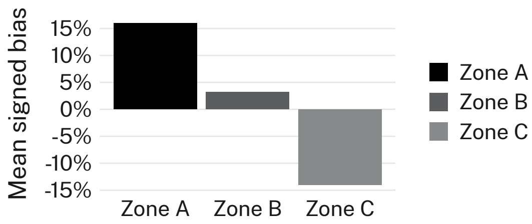


Figure 2: Mean signed bias in the assembly headcount relative to the swipe-derived figure, by zone: Zone A over-counts by 16.0%, Zone C under-counts by 14.0%, and Zone B sits closest to parity at 3.2%.

Applying Facilities and Safety's one-warden-per-twelve-occupants formula to the trailing three-drill mean of each reading separately, the assembly-count-derived establishment and the swipe-log-derived establishment agree on six of the building's nine floors and diverge by one warden on the remaining three. Two of the three divergences follow the zone bias directly: Level 2 (Zone A) is assigned one additional warden under the assembly count, and Level 7 (Zone C) one fewer. The third, Level 5 (Zone B), diverges in the other direction — its swipe-derived mean crosses the twelve-person rounding threshold that its assembly mean does not — despite Zone B carrying, on average, the smaller of the two readings' disagreements.

Beginning at drill five, marshals received a fifteen-minute refresher briefing on tally procedure. Figure 3 compares each floor-drill's absolute bias for drills 1-4 against drills 5-7: mean absolute bias moves from 25.4% to 20.9% (Mann-Whitney $U = 404$, $z = -1.14$, $p = 0.26$, n.s.), and the Zone A / Zone C asymmetry is unchanged in direction across both periods; we retain the refresher in the standard drill protocol for completeness.

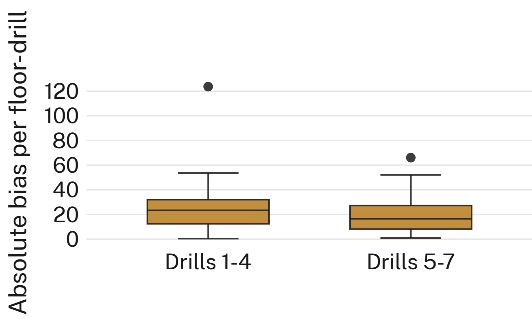


Figure 3: Per-floor-drill absolute bias before and after the marshal refresher briefing introduced at drill five: the distributions are statistically indistinguishable (Mann-Whitney $p = 0.26$, n.s.).

## Discussion

These results suggest that a marshal-reported assembly headcount, though procedurally complete and available within the hour a drill finishes, tracks the same alarm's swipe-derived occupancy only moderately, and that the disagreement is structured by the building's own zone-and-stairwell geography rather than distributed as noise around an otherwise reliable reading. The refresher briefing's null effect on bias magnitude may indicate the disagreement originates in the assembly procedure's zone boundaries and stair-descent timing rather than in individual marshal error, a possibility consistent with the building's own layout rather than a training gap. Because the current establishment formula treats the two readings as interchangeable, three of the nine floors carry a warden allocation that a swipe-log-derived formula would set differently; whether that divergence matters for evacuation safety on those floors is a question this study is not positioned to answer, though it may be a modest input to a future review of the establishment formula. We do not take these findings to indicate the assembly-point procedure is failing at its primary purpose, which remains verifying that a floor has fully evacuated; the divergence concerns only the headcount's second, unaudited use as a staffing input.

## Limitations

Our sample of seven drills and nine floors over a single term is modest, though the consistency of the Zone A / Zone C asymmetry across all seven drills suggests the pattern is not an artefact of any one drill's conditions. The swipe-derived occupancy figure assumes every floor occupant badges in and out at their own floor's reader, which may itself undercount visitors or occupants passing through via another floor's reader; if anything, this would narrow rather than widen the disagreement we report, since it would push the swipe figure toward, not away from, the assembly count. The single building studied precludes claims about whether the zone-and-stairwell pattern generalises to other campus buildings' assembly procedures, a question we consider a natural next step given the present, already consistent, findings.

## Conclusion

We compared a nine-storey administration building's fire-drill assembly-point headcount against its own swipe-card system's occupancy figure for the same alarm across seven drills and nine floors, finding the two readings correlate only moderately and diverge in a direction and magnitude specific to each assembly zone. A marshal refresher briefing left the disagreement unchanged. Under the establishment formula currently in force, three of the building's nine floors carry a warden allocation the swipe-derived figure would set differently. We suggest that occupancy proxies drawn from safety-compliance procedures be validated against an available passive log before they are extended to a second, staffing-facing use. Future work should test whether the same zone-boundary and stair-descent pattern recurs across the University's other multi-storey buildings, and whether an occupancy-linked establishment formula that blends both readings narrows the divergence reported here.

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